Unit 4 General Principles of Chemistry I – Rates, Equilibria and Further Organic Chemistry

IA2 compulsory unit

Externally assessed

4.1 Unit description

Introduction

In this unit students make a quantitative study of chemical kinetics and take further their study of organic reaction mechanisms.

The topics of entropy and equilibria show how chemists are able to predict quantitatively the direction and extent of chemical change.

The organic chemistry in this unit covers carbonyl compounds, plus carboxylic acids and their derivatives.

Students are required to apply their knowledge gained in Units 1 and 2, to all aspects of this unit. This includes nomenclature, ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balance equations and calculate chemical quantities.

How chemists work

Through practical work, students will learn about the methods used to measure reaction rates. They will collect data, analyse it and interpret the results. They then see how a knowledge of rate equations and other evidence can enable chemists to propose models to describe the mechanisms of reactions.

The study of entropy introduces students to the methods of thermodynamics and shows how chemists use formal, quantitative and abstract thinking to answer fundamental questions about the stability of chemicals and the direction of chemical change.

The unit tests the equilibrium law by showing the degree to which it can accurately predict changes during acid-base reactions, notably the changes to pH during titrations.

The historical development of theories explaining acids and bases shows how scientific ideas change as a result of new evidence and fresh thinking.

Chemistry in action

This unit shows how the principles of kinetics and thermodynamics can help to achieve optimal conditions for the manufacture of chemicals.

The study of buffer solutions shows the varied importance of equilibrium systems in living cells, in medicines, in foods and in the natural environment.

The two broad areas of application of chemistry are synthesis and analysis. In this unit synthesis is illustrated by reactions of carbonyl compounds (notably with cyanide ions) and the production of esters for use as solvents, flavourings and perfumes. The main analytical technique featured is nmr including coverage of magnetic resonance imaging.

Core practicals

The following specification points are core practicals within this unit that students should complete:

4.3c

4.3e

4.4g

4.8.2c

4.8.3d

4.8.4b

4.8.4c

These practicals may appear in the written examination for Unit 4.

Use of examples

Examples in practicals

Where 'e.g.' follows a type of experiment in the specification students are not expected to have carried out that specific experiment.

However they should be able to use data from that or similar experiments.

For instance in this unit, *4.3g How fast? – rates*, the specification states:

investigate the activation energy of a reaction, e.g. oxidation of iodide ions by iodate(V).

Students will be expected to have investigated the activation energy of a reaction, but they may or may not have done this by the oxidation of iodide ions by iodate(V).

In the unit test students could be given experimental data for this or any other reaction, and be expected to use this data to evaluate or estimate the activation energy.

Examples in unit content

Where 'e.g.' follows a concept students are not expected to have been taught the particular example given in the specification. They should be able to illustrate their answer with an example of their choice.

For instance in this unit, *4.7m Acid/base equilibria*, the specification states:

explain the importance of buffer solutions in biological environments, e.g. buffers in cells and in blood (H_2CO_3/HCO_3^-) and in foods to prevent deterioration due to pH change (caused by bacterial or fungal activity).

Students will be expected to explain the importance of buffer solutions in biological systems, but they may or may not have looked at buffers in cells and blood, or in food.

In the unit test students could be asked to illustrate the importance of buffer solutions with a biological example that they select themselves. This could be one listed as an example or it could be another example.